

## Supplementary

### 1. Attention Analysis of molecular prediction task

In the main manuscript, we validated Token-Mol's robust predictive capabilities in molecular property prediction tasks. While predictive accuracy is essential for scientific research, it is equally crucial for a model to validate scientific hypotheses. To this end, we focused on drug solubility and acute oral toxicity, areas extensively explored and significant for verifying model interpretability<sup>1</sup>. Using the ESOL and LD50 datasets, we illustrated how Token-Mol can extract relevant features from specific atoms or functional groups and assign corresponding attention weights to related tokens.

The ESOL dataset relates to molecular solubility, where various functional groups significantly influence solubility. Molecules containing polar functional groups, such as hydroxyl, carboxyl, and amino groups, generally exhibit higher solubility<sup>2</sup> due to their ability to form hydrogen bonds with water molecules. In contrast, hydrophobic functional groups, such as alkyl and halogen groups, tend to reduce solubility. Supplementary Figure 6a shows that in hydrophilic substructures, sp<sup>2</sup> hybridized carbon, hydroxyl groups, and amine groups substantially impact the molecule's attention weights. For low-solubility molecules, the figure indicates that chlorophenyl and halogen atoms receive high attention weights. Moreover, a comparison in Supplementary Figure 6a reveals that, unlike high-solubility molecules where single functional groups often have significant influence, in drug design, the hydrophobicity of a molecule typically has a more holistic impact, with modifications of key hydrophobic groups affecting the overall structure's hydrophobic properties.

In drug toxicity research, we typically focus on toxicophores, or structural alerts (SAs)<sup>3</sup>, defined as key substructures responsible for specific toxicities and commonly used to elucidate these relationships. For example, in the LD50 dataset, we identified the nitrosamide and phosphoric trimester toxicophores as reported by Wu et al.<sup>4</sup>. Nitrosamide is a frequently occurring SA in both mutagenic and non-mutagenic compounds<sup>5</sup>, while phosphoric trimester is commonly found in compounds with acute oral toxicity<sup>6</sup>. We filtered the dataset for molecules containing nitrosamide and

phosphoric trimester toxicophores and visualized the model's attention weights for these molecules. As shown in Supplementary Figure 6b, the model indeed assigns high attention weights to these toxicophores in all molecules containing the identified toxic fragments.

## 2. Datasets Details

The descriptions of various benchmarks dataset are listed in Supplementary Table 6.

- BACE is a database that catalogues molecules with 2D structures and properties, focusing on inhibitors of human  $\beta$ -secretase 1. It provides qualitative (binary label) binding results for these inhibitors.
- The BBBP dataset comprises molecules with measured permeability properties, specifically assessing their ability to penetrate the blood-brain barrier.
- ClinTox is a dataset that encompasses both FDA-approved drugs and compounds that have been discontinued from clinical trials due to toxicity concerns.
- SIDER is a database that catalogs marketed drugs along with their associated adverse drug reactions (ADRs), categorized into 27 system organ classes.
- Tox21 assesses the toxicity of compounds across 12 different targets, spanning nuclear receptors and stress response pathways. It served as a public database for the 2014 Tox21 Data Challenge.
- ToxCast comprises toxicology data for thousands of molecules, offering multiple toxicity labels derived from high-throughput screening experiments conducted on a vast chemical library.
- The ESOL (Free Solvation Energy of Neutral Organic Molecules in Water) dataset is a standard dataset used in machine learning and cheminformatics research. It aims to predict the free solvation energy of organic molecules in water.
- The Free Solvation Database, FreeSolv (SAMPL), provides experimental and calculated hydration free energy of small molecules in water. The calculated values are derived from alchemical free energy calculations using molecular dynamics simulations.
- Lipophilicity measures the ability of a drug to dissolve in a lipid (e.g. fats, oils)

environment. High lipophilicity often leads to high rate of metabolism, poor solubility, high turn-over, and low absorption.

- The human colon epithelial cancer cell line, Caco-2, is used as an in vitro model to simulate the human intestinal tissue. The experimental result on the rate of drug passing through the Caco-2 cells can approximate the rate at which the drug permeates through the human intestinal tissue.
- Aqueous solubility measures a drug's ability to dissolve in water. Poor water solubility could lead to slow drug absorptions, inadequate bioavailability and even induce toxicity. More than 40% of new chemical entities are not soluble.
- Acute toxicity LD50 measures the most conservative dose that can lead to lethal adverse effects. The higher the dose, the more lethal of a drug.

### 3. Different decoding strategy in pocket-based generation

In generative language models, different decoding strategies (also known as sampling strategies) can significantly affect the quality of the generated sequences. In pocket-based generation tasks, the input pocket information functions similarly to a prompt. To enable the model to generate more diverse sequences under the same prompt, we employed temperature sampling as the decoding strategy. In contrast, TamGen utilizes beam search as its sampling method.

Our analysis in the main manuscript revealed that the molecules generated by TamGen exhibited significantly lower SA Score compared to those from Token-Mol. In addition to an extra VAE as encoder to encode the original ligands, we believe that the choice of sampling strategy of TamGen may also influence the generated sequences, as reflected in the SA Score. Furthermore, we introduced a new sampling strategy by implementing top-k sampling as an alternative for Token-Mol in pocket-based generation tasks, which aligns more closely with the sampling methods of GPT-like models. For this, we set top-k=10 and top-p=0.95.

The generated molecular metrics confirm our hypothesis (Supplementary Figure 9 & Supplementary Table 10). The use of top-k sampling with Token-Mol does not significantly differ from temperature sampling in terms of the quantity and validity of

generated sequences. Moreover, there are no notable changes in similarity as well as the distributions of affinity and most molecular properties. Notably, molecules generated using top-k sampling exhibit significantly lower SA Score, approaching or even falling below those of TamGen. Consequently, this leads to a decline in the higher score which reflects high affinity enrichment, alongside a marked increase in LogP.

Analyzing from the molecular structure perspective (Supplementary Table 11), we find that top-k sampling results do not differ significantly from those of temperature sampling concerning most structural features, such as the average number of rings or chiral centers. However, there are variations in the proportions of macrocyclic structures and fused ring systems in the top-k sampling results. These characteristics are closely related to the calculation of the SA score. Additionally, the number of substructure features extracted using the ECFP4 is more aligned with TamGen in top-k sampling results.

Based on these findings, we hypothesize that top-k as a relatively “conservative” sampling strategy, reduces the generation of tokens with low occurrence probabilities, resulting in decreased substructure diversity and a tendency to generate lipophilic fragments which are more common. This reduction in SA Score is accompanied by an increase in lipophilicity. However, these fragments do not provide strong affinity contributions when binding into pocket, leading to a decrease in high affinity enrichment.

#### **4. Validity and uniqueness during reinforcement learning process**

To demonstrate the effectiveness of the reinforcement learning process, we conducted additional analyses to evaluate the efficacy and uniqueness of the generated compounds (Supplementary Figure 10a). In terms of validity, we observed that for both the ARA2A and CDK2 targets, the validity of the generated molecules fluctuated around 0.8 during the initial reinforcement learning steps. As training stabilized, this validity gradually converged to approximately 0.9. This finding indicates that the majority of sequences generated throughout the reinforcement learning process were valid, rather than merely a small number of effective compounds artificially boosting scores. This outcome supports the effectiveness of our reward function design, which includes a penalty for

invalid molecules.

Regarding uniqueness (hereafter referred to as Uni), we approached this analysis in two parts. First, using the traditional analysis of molecular generation (Brown et al., 2019), we calculated Uni based on the canonical SMILES of the generated molecules (SMILES Uni). Our findings showed a significant decline in this metric as the reinforcement learning progressed, indicating a noticeable decrease in molecular diversity, with the final steps potentially yielding only 1-2 unique molecules. This is consistent with our expectations. Since we also generate torsional angle, so we considered the overall sequences uniqueness (Seqs Uni), which maintained a value of 1 throughout the entire learning process, indicating that none of the generated sequences were duplicated.

Although the Seq Uni consistently remains at 1, the conformations of the generated molecules do not exhibit significant randomness due to the diversity of torsion angles. Instead, they are concentrated in one or several main conformational spaces. We calculated the RMSD matrix between all conformations of the molecules with the highest number of conformations at each step after 400 steps, which corresponds to the convergence of SMILES Uni, and performed hierarchical clustering using the UPGMA algorithm from the SciPy library, setting the RMSD threshold to 1 Å. After obtaining the hierarchical clustering results, we normalized the frequency of conformations within each cluster and sorted them in descending order to highlight clusters containing more molecules. We selected the top 5 frequencies from each group (using 0 if fewer than 5 clusters were obtained) and created a heatmap as shown in Supplementary Figure 10b.

The results indicate that in both targets, after reinforcement learning process led the SMILES portion of the generated sequences to approach local optima, the torsion angles are observed to be maintained within 1 to 3 main conformational spaces. It is particularly evident in ARA2A, where the majority of generated conformations clustered into a single conformational space. This phenomenon is ideal for pocket-based molecular generation, as it can be seen as the optimization of the 3D conformations of generated molecules through reinforcement learning, resulting in a small set of stable conformations that are well-suited for binding within the pocket. The diverse range of actual conformations can be viewed as minor perturbations around these primary conformations, remaining

within the acceptable bounds of binding kinetics. This aligns with our previously mentioned goal of using reinforcement learning to ensure that both SMILES and three-dimensional conformations converge to local optima, thereby obtaining candidate molecules with optimal performance.

#### **5. Stability of conformational information during the reinforcement learning process.**

Due to the absence of specific constraints on conformational information in the reward function we designed, and given the significant role that 3D conformations play in the generation task, it is equally important to pay attention to the variations in conformational information during the reinforcement learning process. Otherwise, molecules with unreasonable conformations may still achieve high affinity through optimization by docking tools, thereby misleading the reward function and resulting in reinforcement learning outcomes that deviate from the ideal optimization results. We analyzed this issue from the perspectives of torsional angles and conformational rationality.

We conducted an analysis of the torsional angle distribution (Supplementary Figure 11a). As stated in the main text, we utilized Jensen-Shannon (JS) divergence to quantify this metric. The reference molecules were taken as those generated during the initial step of reinforcement learning, which should have a torsional angle distribution consistent with that produced by the prior model. We selected specific torsional angles present in each reinforcement learning process and monitored the JS divergence of these distributions. The results indicate that the JS divergence experienced a significant increase from an initial value of 0, followed by rapid fluctuations. However, it quickly converged into a more stable range. This suggests that in the early stages of reinforcement learning, as the policy evolves, the torsional angles of the generated molecules also change rapidly. In later stages, stability is achieved as the reward function converges.

Despite this stability, we observed that some torsional angles exhibited notably high JS divergence, exceeding 0.5, indicating that there were indeed some changes in torsional angle information compared to the prior model before training. During the optimization of model parameters via reinforcement learning, the strategies for generating torsion

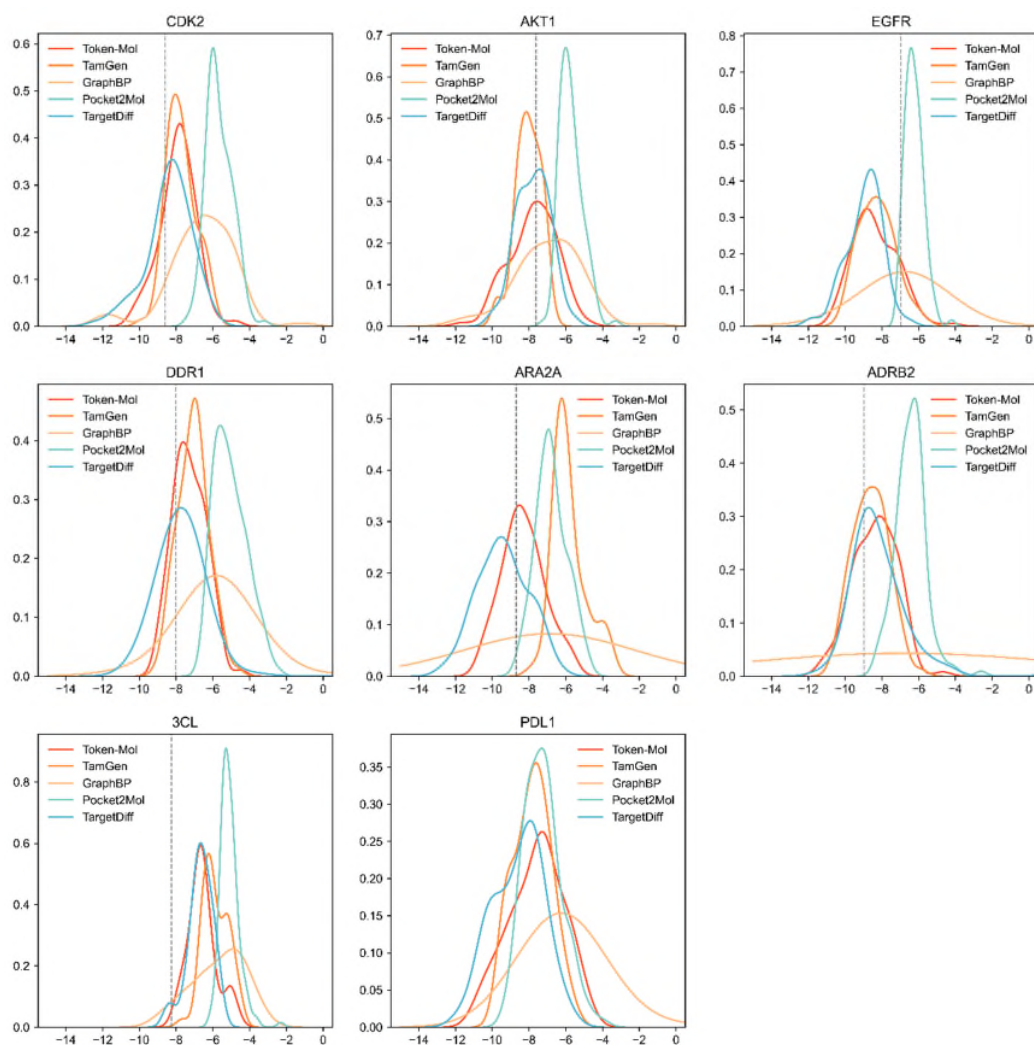
angles are altered. This is due to the absence of constraints within the reinforcement learning rewards that would enforce consistency with prior torsion angle distributions. Such variations in torsion angle distributions are acceptable, provided they do not compromise the integrity of the stereochemical conformations.

To further validate whether the distribution of torsion angles remains within a normal range and ensure the validity of the generated conformations, we compared it with the distribution of the CrossDocked2020 training set as a reference. The molecular conformations in this dataset are derived from crystallographic data and have been optimized using the UFF force field. Furthermore, all selected torsion angles exist in at least 1,124 conformations within this dataset, making its distribution reflective of the real-world distribution of torsion angles. We also calculated the JS divergence between the molecules at each step of the RL process and the reference set (Supplementary Figure 11b). The results indicate that these torsion angles exhibit a JS divergence ranging from 0.2 to 0.3 during the RL process. Considering that the number of conformations available at each step of the RL process is significantly smaller than that of the reference set, this range suggests that the overall distribution of torsion angles during the RL training process does not deviate significantly from the reference distribution. This finding demonstrates the conformational validity of the generated structures throughout the RL process.

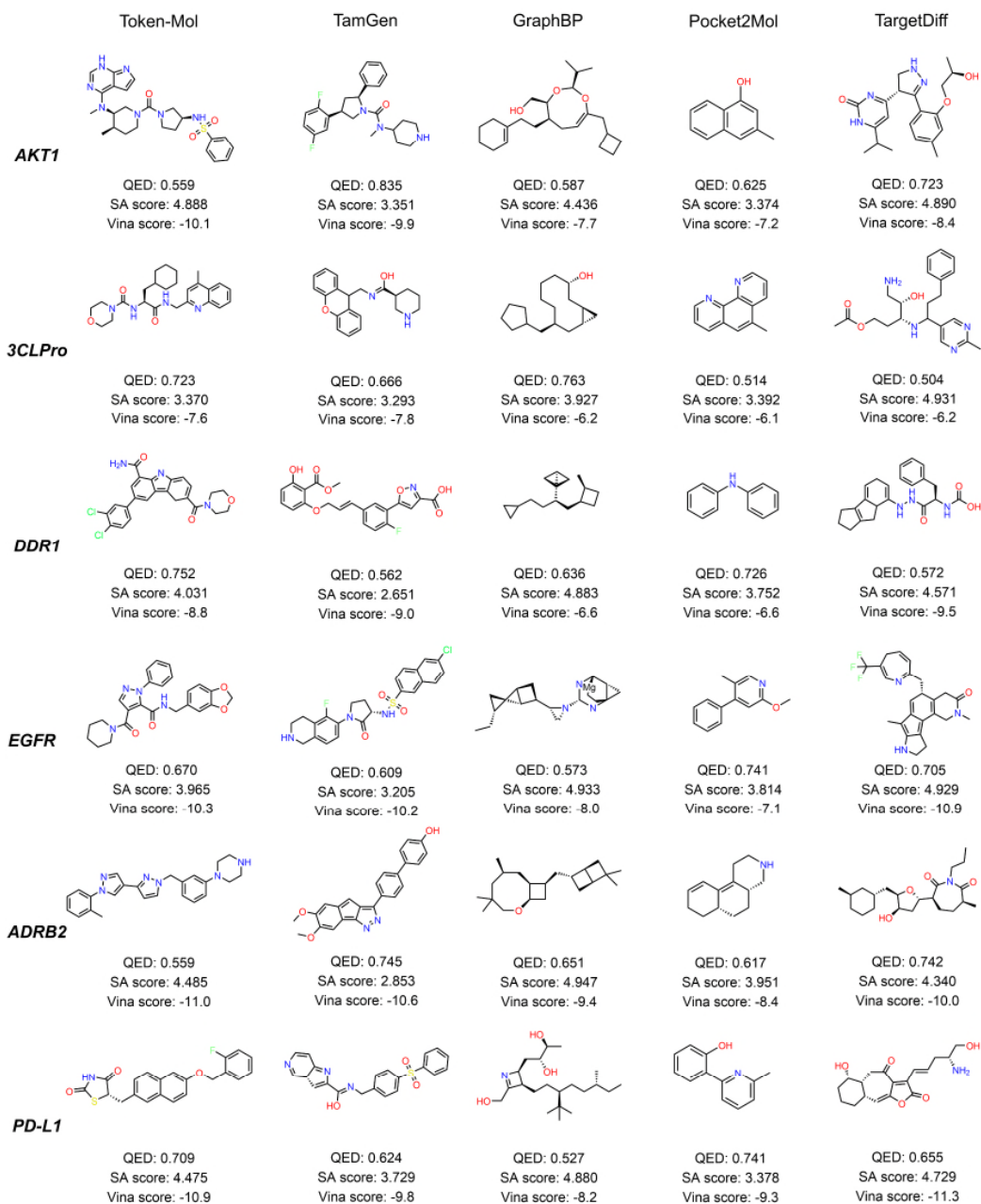
Moreover, we employed PoseBusters (Buttenschoen et al., Chem. Sci. 2023) to analyze the rationality of the stereo conformations (Supplementary Figure 11c). We selected indicators related to molecular conformations and internal energy. The results showed that for the ARA2A target, the pass rate of these indicators reached 1 in the latter stages of training, indicating that the generated molecular conformations were highly rational. For CDK2, while the internal energy pass rate did not converge to 1, the other indicators remained within the range of 1, and the internal energy metrics stabilized at a commendable range of 0.6-0.8. This evidence suggests that, even in the absence of specific constraints on 3D information within the reward function, the generation of unreasonable conformations was minimal. Although the 3D information underwent some transformations, it did not lead to a complete loss as the policy altered.



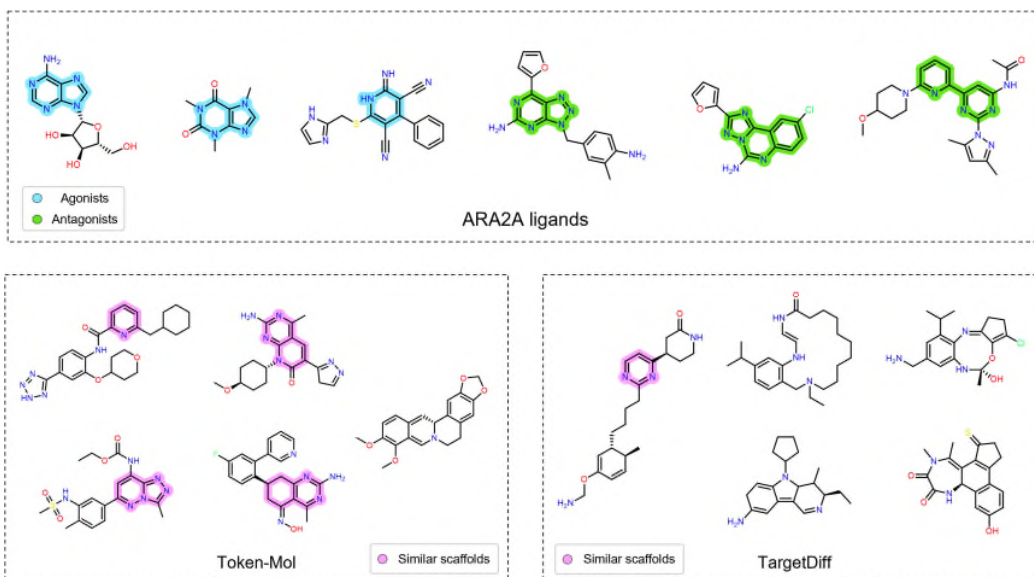




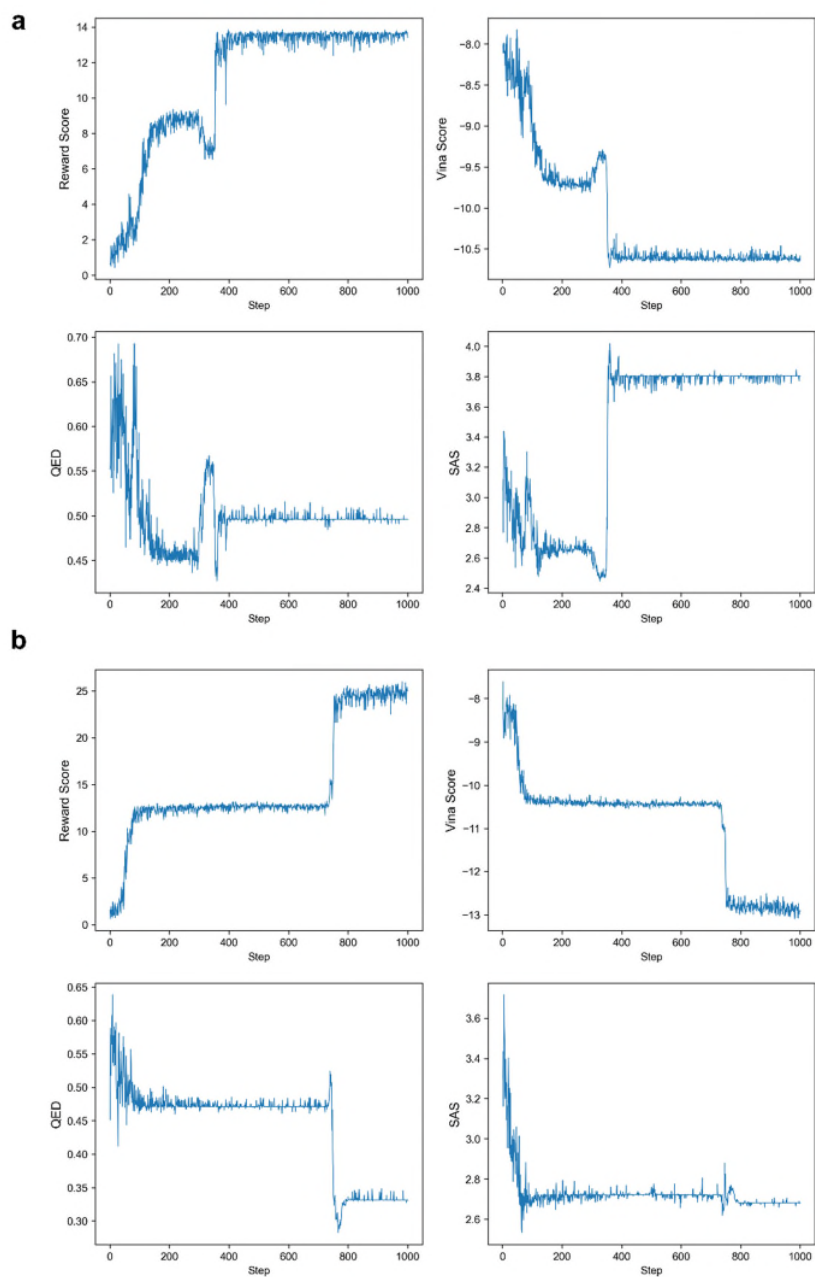
**Supplementary Figure 2 | Vina score distributions of Token-Mol and baseline models on the selected real-world targets.** The gray dashed line and the area to its left represent higher affinity molecules. Source data are provided as a Source Data file.



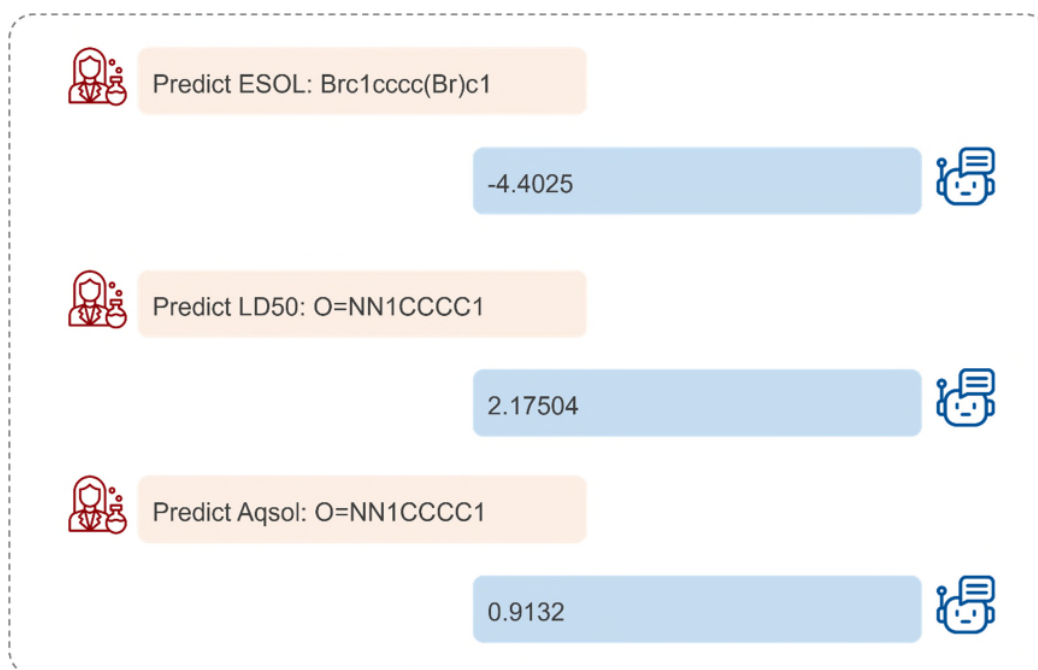
Supplementary Figure 3 | The ‘drug-like’ molecules with the highest affinity and their properties on the selected real-world targets which are not represented in the main text.



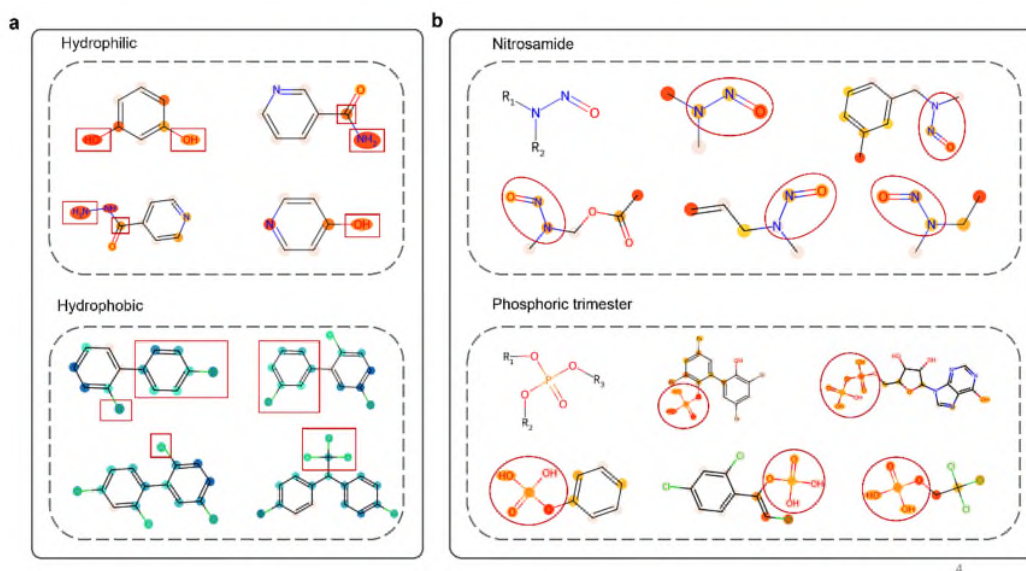
**Supplementary Figure 4 | Comparison between the real-world ligands of ARA2A and molecules with generated by Token-Mol and TargetDiff.** The scaffolds of the real-world ARA2A ligands and those of molecules similar to the real-world ligands generated by the models are highlighted.



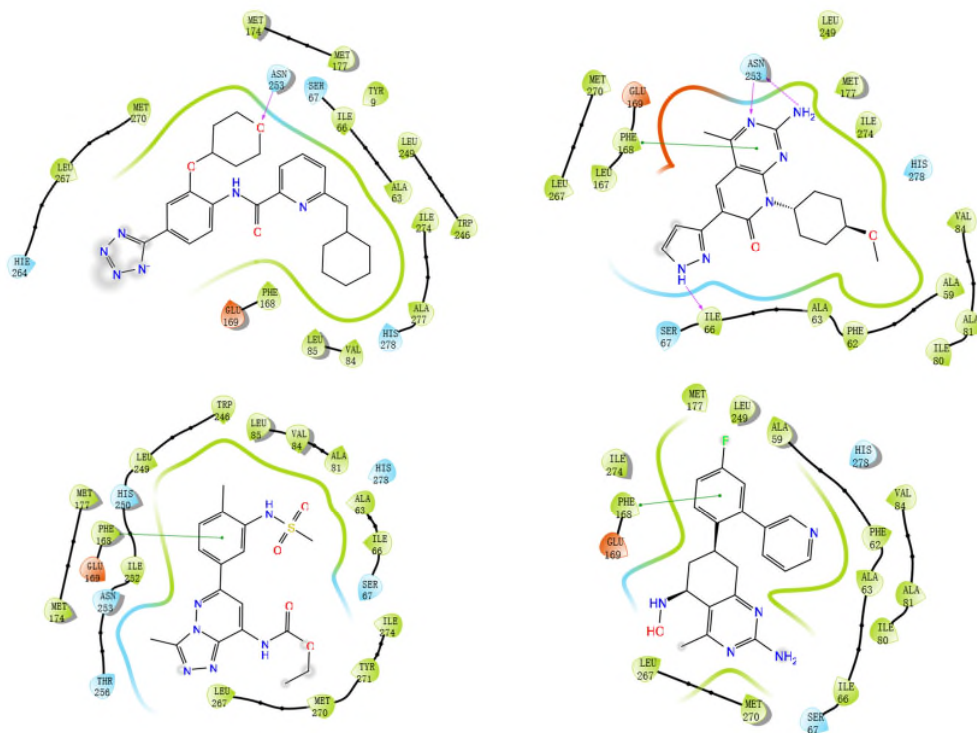
Supplementary Figure 5 | Metrics during the reinforcement learning to optimized the binding affinity without any constraints in (a) CDK2 and (b) EGFR.



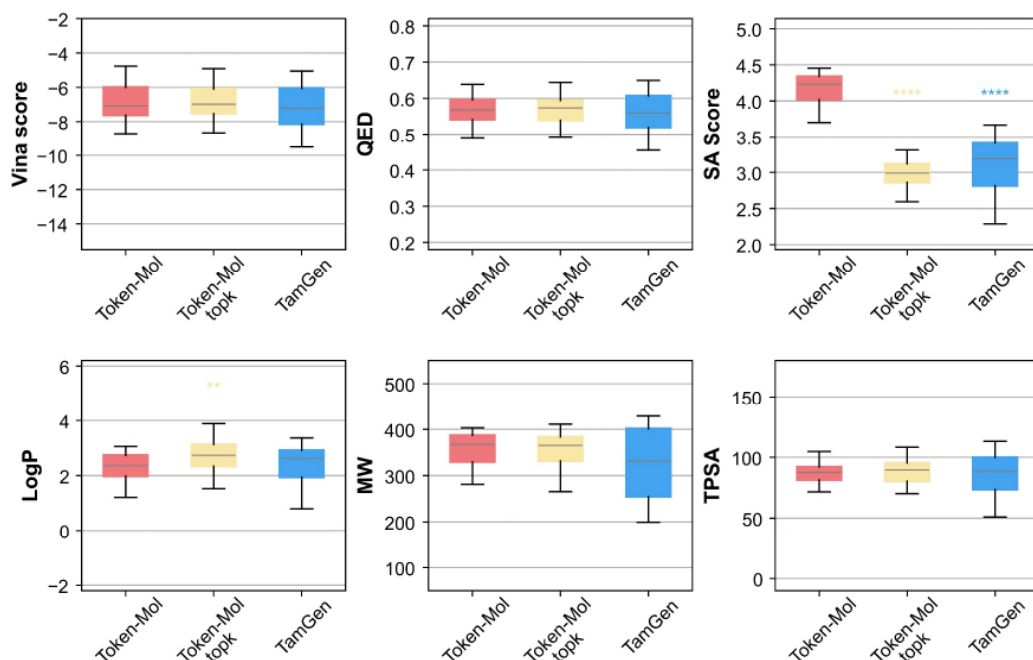
Supplementary Figure 6 | The example of “Chat to Token-Mol”.



Supplementary Figure 7 | Visualization of the attention weights for different molecules. (a) The attribution visualization of aqueous solubility. The darker the color, the higher the weight. (b) Nitrosamide and phosphoric trimester of acute oral toxicity.

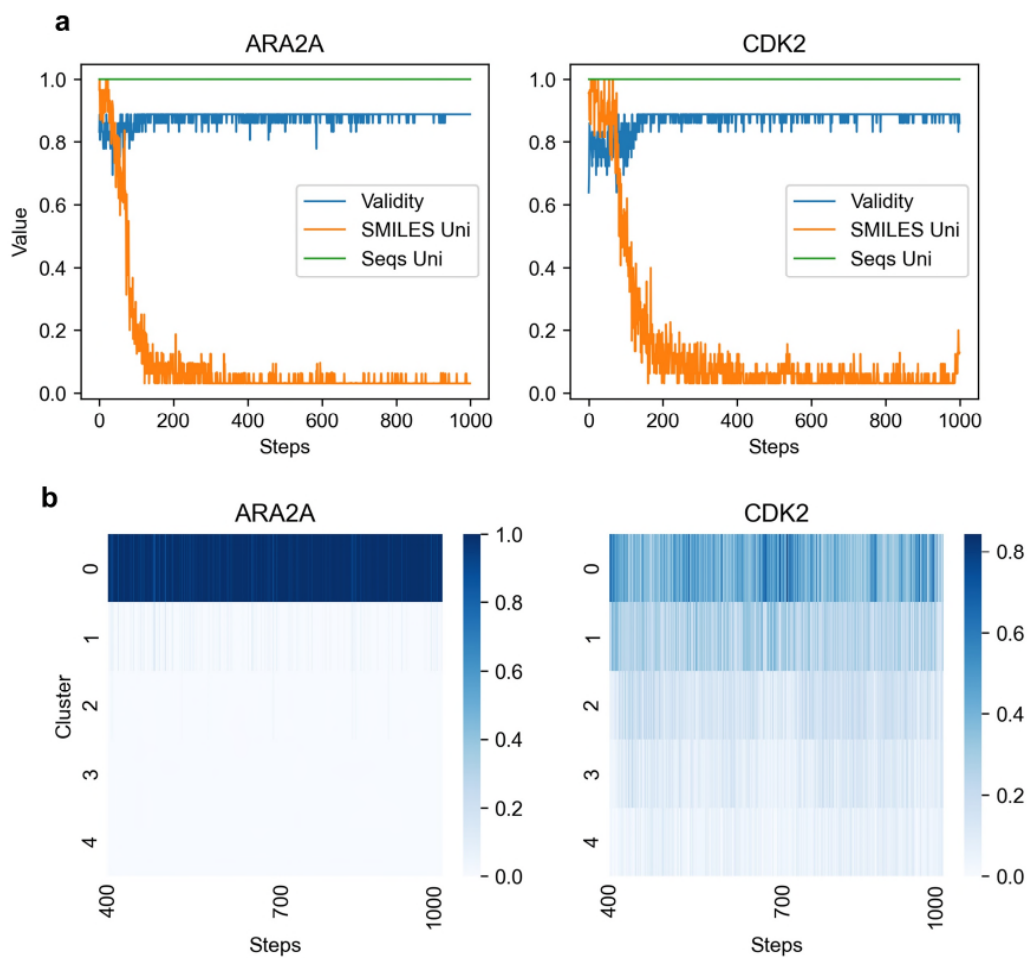


Supplementary Figure 8 | The 2D interaction diagrams of the four generated molecules with similar scaffolds to real-world ligands with ARA2R. The docking poses were obtained using Qvina2, and the interaction diagrams were drawn in Maestro's 2D Sketcher.



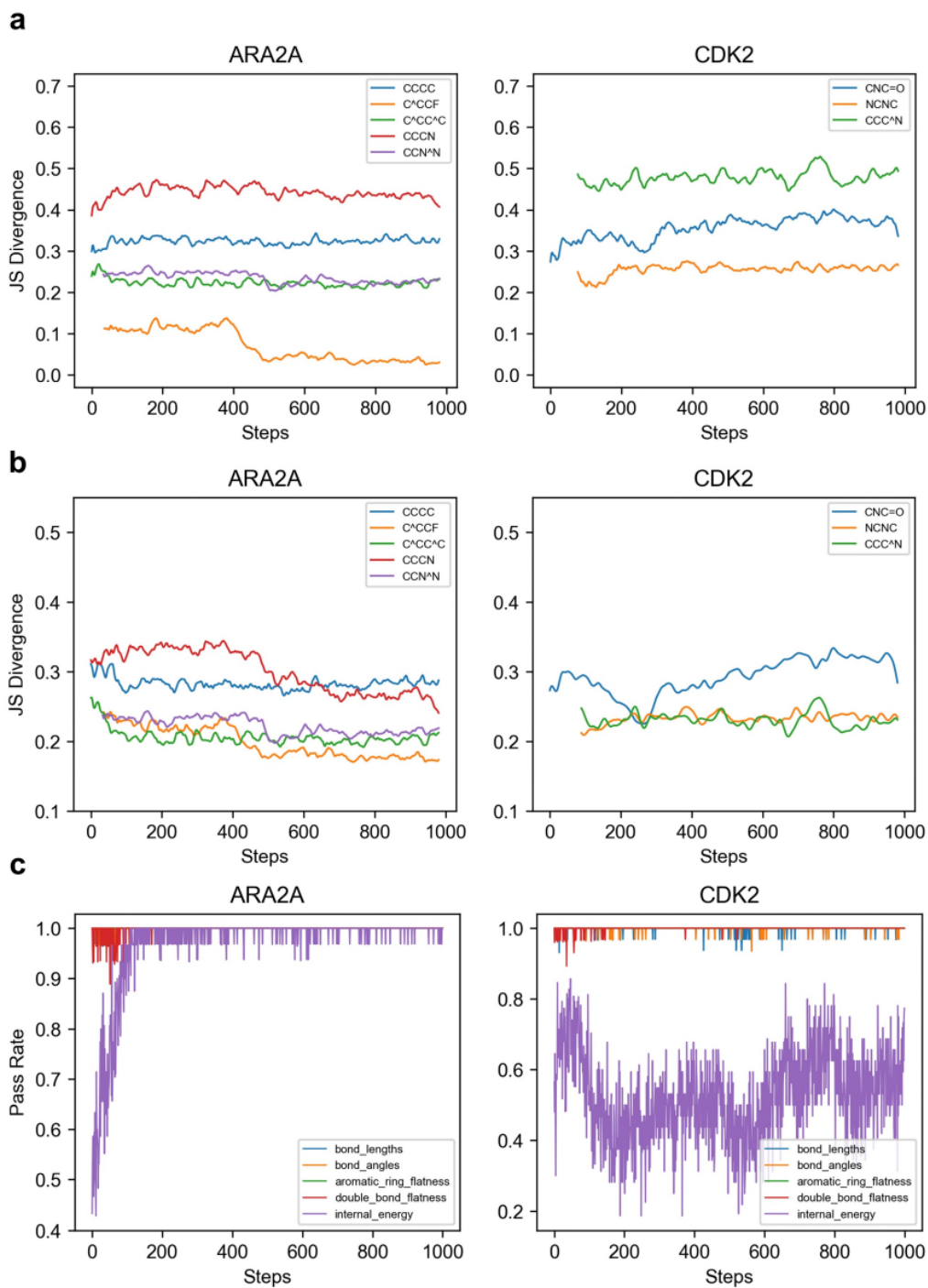
**Supplementary Figure 9 | Distribution of average molecular properties of generated molecules within the test set (n=77) from Token-Mol in different decoding strategy and TamGen.** The data are shown with box plots which display the median (center line), upper and lower quartiles (box limits), 1.5x interquartile range (whiskers). Inter-group differences were analyzed using a two-sided Mann-Whitney U test, with p-values adjusted for multiple comparisons using the Bonferroni correction. Other methods are described in the Methods section of the main text. All experimental groups were independently compared to the control group (Token-Mol), with a significance level set at  $\alpha = 0.05$ . Detail p-values of each group can be found at Supplementary Table 14. \*p < 0.05, \*\*p < 0.01, \*\*\*p < 0.001, \*\*\*\*p < 0.0001 vs control. Source data are provided as a Source Data file.





**Supplementary Figure 10** | (a) Monitoring of validity and uniqueness during reinforcement learning. (b) Hierarchical clustering heatmap based on RMSD. Source data are provided as a Source Data file.





**Supplementary Figure 11** | (a) Monitoring of JSD value with the initial step as reference during reinforcement learning. (b) Monitoring of JSD value with CrossDocked2020 training set as reference during reinforcement learning. The curves in a and b are smoothed using the Gaussian filter from the scipy library, with  $\sigma = 2$ . (c) Monitoring of conformational rationality during reinforcement learning via PoseBusters. Source data are provided as a Source Data file.

Supplementary Table 1 | Detailed data about the molecular properties of molecules generated in the test set by Token-Mol and other baseline models.

| Metric       |      | Ori.    | Token-Mol    | TamGen       | GraphBP | Pocket2Mol    | TargetDiff    |
|--------------|------|---------|--------------|--------------|---------|---------------|---------------|
| Vina score ↓ | Avg. | 7.010   | -6.862       | -7.153       | -6.659  | <b>-7.729</b> | -7.236        |
|              | Med. | -6.900  | -7.077       | -7.211       | -6.733  | -7.071        | <b>-7.285</b> |
| QED ↑        | Avg. | 0.492   | <b>0.568</b> | 0.558        | 0.511   | 0.536         | 0.493         |
|              | Med. | 0.470   | <b>0.568</b> | 0.559        | 0.510   | 0.561         | 0.493         |
| SA score ↓   | Avg. | 4.214   | 4.166        | <b>3.072</b> | 5.271   | 5.846         | 4.858         |
|              | Med. | 4.277   | 4.232        | <b>3.191</b> | 5.301   | 5.716         | 5.029         |
| Lipinski ↑   | Avg. | 4.247   | 4.666        | <b>4.887</b> | 4.749   | 4.782         | 4.586         |
|              | Med. | 5.000   | 4.684        | 4.910        | 4.746   | <b>4.958</b>  | 4.776         |
| LogP         | Avg. | 1.002   | 2.319        | 2.389        | 3.254   | 1.706         | 1.446         |
|              | Med. | 1.286   | 2.363        | 2.623        | 2.441   | 1.145         | 1.342         |
| MW           | Avg. | 337.341 | 355.414      | 326.963      | 257.307 | 274.273       | 329.241       |
|              | Med. | 326.013 | 366.857      | 330.929      | 257.772 | 242.405       | 340.350       |
| TPSA         | Avg. | 114.292 | 87.320       | 86.260       | 39.834  | 72.246        | 94.945        |
|              | Med. | 87.300  | 87.660       | 88.490       | 39.811  | 67.345        | 88.997        |

Supplementary Table 2 | Jensen-Shannon divergence between the test set and molecules generated by Token-Mol and baseline models.

| Jenson-Shannon Divergence ( $\times 10^{-1}$ ) ↓ |              |        |         |            |            |
|--------------------------------------------------|--------------|--------|---------|------------|------------|
|                                                  | Token-Mol    | TamGen | GraphBP | Pocket2Mol | TargetDiff |
| CNCC                                             | <b>0.240</b> | 1.335  | N/A     | 1.749      | 1.637      |
| C^CCC                                            | <b>0.281</b> | 1.178  | N/A     | 1.385      | 1.496      |
| CCCC                                             | <b>0.326</b> | 1.472  | N/A     | 1.672      | 1.675      |
| NCCC                                             | <b>0.421</b> | 1.457  | N/A     | 2.269      | 1.767      |
| NCC^C                                            | <b>0.427</b> | 1.224  | N/A     | 2.173      | 1.644      |
| COC^C                                            | <b>0.432</b> | 1.212  | N/A     | 1.372      | 1.545      |
| CCC=O                                            | <b>0.336</b> | 1.347  | N/A     | 1.496      | 1.547      |
| CNC=O                                            | <b>0.405</b> | 1.433  | N/A     | 1.517      | 1.581      |
| CNC^C                                            | <b>0.393</b> | 1.232  | N/A     | 1.676      | 1.406      |
| C^CC^C                                           | <b>0.552</b> | 1.175  | N/A     | 1.397      | N/A        |
| Avg.                                             | <b>0.381</b> | 1.306  | N/A     | 1.671      | 1.589      |

N/A means this type of torsion angle did not appear in the molecules or is not detected due to the unreasonable conformation of the molecule. Symbol in the bond types: “=” represents double bond, “^” represents aromatic bond.

**Supplementary Table 3 | Structural information of generated molecules from each model.**

| Metric                   | Token-Mol | TamGen | GraphBP | Pocket2Mol | TargetDiff |
|--------------------------|-----------|--------|---------|------------|------------|
| Chiral centers           | 1.273     | 1.011  | 5.784   | 0.917      | 3.553      |
| Torsion angles           | 6.386     | 5.118  | 5.175   | 2.727      | 4.943      |
| Rotatable bonds          | 5.326     | 4.225  | 4.983   | 2.359      | 4.372      |
| Ring numbers             | 2.846     | 2.887  | 3.574   | 3.361      | 2.674      |
| Largest ring             | 5.573     | 5.883  | 5.176   | 5.963      | 6.270      |
| Extremely large ring (%) | 0.775     | 0.404  | 13.803  | 3.073      | 14.025     |
| Fused rings (%)          | 47.830    | 47.604 | 65.734  | 70.156     | 54.279     |
| Fused system size        | 2.314     | 2.338  | 3.230   | 3.653      | 2.916      |
| Total fragments          | 21,408    | 18,395 | 53,545  | 56,132     | 74,635     |

Chiral centers/Torsion angles/Rotatable bonds/Ring numbers: the number of chiral centers/torsion angles/rotatable bonds/rings in the molecules; Largest ring: atom number of the largest ring in the molecules; Extremely large ring: the ratio of molecules obtain ring with atom number higher than 7; Fused rings: the ratio of molecules with fused ring system; Fused system size: the number of rings within the fused ring system. Total fragments: Total number of non-redundant substructure features (fragments) extracted based on ECFP4.

Supplementary Table 4 | The rate of higher affinity molecules and the average QED values of generated molecules from different models on the selected real-world targets.

| Higher Affinity (%) # |              |               |         |              |               |
|-----------------------|--------------|---------------|---------|--------------|---------------|
| Targets               | Token-Mol    | TamGen        | GraphBP | Pocket2Mol   | TargetDiff    |
| CDK2                  | 21.591       | 13.542        | 8.235   | 0            | <b>38.043</b> |
| AKT1                  | 52.874       | <b>67.010</b> | 34.118  | 0            | 52.174        |
| EGFR                  | 58.559       | 51.220        | 12.048  | 0            | <b>74.359</b> |
| DDR1                  | 18.519       | 14.118        | 6.818   | 0            | <b>38.462</b> |
| ARA2A                 | 40.909       | 0             | 11.765  | 0.917        | <b>73.684</b> |
| ADRB2                 | 31.481       | <b>37.209</b> | 9.091   | 0            | 27.473        |
| 3CLPro                | 27.778       | 2.105         | 20.513  | 0            | <b>28.571</b> |
| PD-L1                 | <b>100</b>   | <b>100</b>    | 96.739  | <b>100</b>   | <b>100</b>    |
| QED                   |              |               |         |              |               |
| CDK2                  | <b>0.630</b> | 0.474         | 0.464   | 0.578        | 0.472         |
| AKT1                  | 0.592        | <b>0.608</b>  | 0.481   | 0.578        | 0.472         |
| EGFR                  | 0.537        | 0.575         | 0.433   | <b>0.614</b> | 0.282         |
| DDR1                  | <b>0.589</b> | 0.586         | 0.430   | 0.585        | 0.331         |
| ARA2A                 | 0.560        | 0.569         | 0.447   | <b>0.628</b> | 0.618         |
| ADRB2                 | 0.571        | 0.551         | 0.478   | <b>0.641</b> | 0.324         |
| 3CLPro                | 0.554        | 0.712         | 0.455   | <b>0.660</b> | 0.247         |
| PD-L1                 | 0.553        | 0.571         | 0.457   | <b>0.620</b> | 0.391         |

# Higher affinity is same as the metric described in the Table 5.

Supplementary Table 5 | Detailed data for drug-like molecules with the highest affinity generated for the two selected targets from different models.

| Target | Model          | Vina Score | QED   | SA Score | Lipinski | TPSA   |
|--------|----------------|------------|-------|----------|----------|--------|
| CDK2   | Token-Mol      | -9.4       | 0.746 | 4.434    | 5        | 92.26  |
|        | Token-Mol (RL) | -10.1      | 0.689 | 3.830    | 5        | 68.10  |
|        | TamGen         | -9.9       | 0.684 | 3.208    | 5        | 58.11  |
|        | GraphBP        | -8.1       | 0.836 | 4.102    | 5        | 37.30  |
|        | Pocket2Mol     | -7.4       | 0.625 | 3.374    | 5        | 20.23  |
|        | TargetDiff     | -8.9       | 0.586 | 4.868    | 5        | 112.74 |
| ARA2A  | Token-Mol      | -9.5       | 0.541 | 4.306    | 5        | 49.69  |
|        | Token-Mol (RL) | -10.3      | 0.701 | 2.567    | 5        | 58.10  |
|        | TamGen         | -7.9       | 0.810 | 2.386    | 5        | 63.32  |
|        | GraphBP        | -8.0       | 0.527 | 4.367    | 5        | 0      |
|        | Pocket2Mol     | -7.1       | 0.741 | 3.814    | 5        | 0      |
|        | TargetDiff     | -11.2      | 0.792 | 4.809    | 5        | 58.89  |

Supplementary Table 6 | The similarity and diversity metrics of the generated molecules in the real-world targets.

|       | Token-Mol           | TamGen        | GraphBP   | Pocket2Mol    | TargetDiff    |
|-------|---------------------|---------------|-----------|---------------|---------------|
|       | Scaffold Similarity |               |           |               |               |
| CDK2  | 0.121               | 0.199         | 0.026     | 0.099         | 0.101         |
| AKT1  | 0.101               | 0.195         | 0.039     | 0.083         | 0.097         |
| 3CL   | 0.103               | 0.161         | 0.050     | 0.083         | 0.086         |
| DDR1  | 0.127               | 0.172         | 0.034     | 0.051         | 0.111         |
| ARA2A | 0.111               | 0.100         | 0.033     | 0.096         | 0.102         |
| ADRB2 | 0.106               | 0.206         | 0.038     | 0.071         | 0.082         |
| EGFR  | 0.110               | 0.188         | 0.037     | 0.090         | 0.102         |
| PD-L1 | 0.106               | 0.203         | 0.029     | 0.072         | 0.094         |
|       | FCD↓                |               |           |               |               |
| CDK2  | <b>42.863</b>       | 43.519        | 52.530    | 47.603        | 46.167        |
| AKT1  | 51.058              | 51.482        | 51.327    | <b>48.625</b> | 53.346        |
| 3CL   | <b>37.741</b>       | 42.012        | 46.988    | 44.441        | 40.010        |
| DDR1  | <b>47.912</b>       | 49.371        | 56.974    | 55.331        | 49.085        |
| ARA2A | <b>53.982</b>       | 57.945        | 64.379    | 61.755        | 54.312        |
| ADRB2 | 43.835              | 44.353        | 48.020    | 47.028        | <b>43.131</b> |
| EGFR  | 66.625              | 64.171        | 68.195    | <b>62.690</b> | 70.251        |
| PDL1  | 47.734              | <b>43.608</b> | 50.706    | 49.660        | 48.367        |
|       | #Circle↑            |               |           |               |               |
| CDK2  | 57                  | 28            | 78        | 36            | <b>90</b>     |
| AKT1  | 46                  | 26            | 79        | 37            | <b>90</b>     |
| 3CL   | 59                  | 15            | <b>73</b> | 45            | 63            |
| DDR1  | 60                  | 39            | 76        | 34            | <b>77</b>     |
| ARA2A | 66                  | 20            | 80        | 28            | <b>95</b>     |
| ADRB2 | 56                  | 31            | 80        | 42            | <b>91</b>     |
| EGFR  | 63                  | 29            | <b>81</b> | 45            | 78            |
| PDL1  | 63                  | 40            | <b>85</b> | 44            | 71            |

Scaffold similarity is the similarity of the Bemis-Murcko scaffolds between reference molecules and generated molecules, calculated with ECFP4 fingerprints. FCD, Fréchet ChemNet Distance. #Circle, a locality-based chemical space coverage measure, derive from the calculation performed on 100 molecules.

**Supplementary Table 7 | Information about the test targets and reference molecules.**

| Target | PDB ID | IC <sub>50</sub> of original ligand                      | Number | QED   | Mean Vina Score |
|--------|--------|----------------------------------------------------------|--------|-------|-----------------|
| CDK2   | 1h00   | 3.5×10 <sup>4</sup> nM                                   | 402    | 0.482 | -8.589          |
| AKT1   | 4gv1   | 4.2 nM                                                   | 156    | 0.461 | -7.612          |
| EGFR   | 2rgp   | 30 nM                                                    | 421    | 0.482 | -8.247          |
| DDR1   | 5bvk   | 1×10 <sup>5</sup> nM                                     | 261    | 0.401 | -8.010          |
| ARA2A  | 3rfm   | >1×10 <sup>4</sup> nM                                    | 3679   | 0.533 | -8.695          |
| ADRB2  | 3ny8   | 2.3 nM                                                   | 457    | 0.405 | -8.977          |
| 3CL    | 7d3i   | 7.3 nM (covalent)<br>1.8×10 <sup>4</sup> nM (Telaprevir) | 448    | 0.438 | -6.966          |
| PD-L1  | 5j89   | ~ 18 nM                                                  | 52     | 0.408 | 4.515           |



**Supplementary Table 8 | Dataset details of molecular property perdition task.**

| Task Type      | Metric  | Dataset                 | Benchmark    | Compounds | Tasks |
|----------------|---------|-------------------------|--------------|-----------|-------|
| Classification | ROC-AUC | BACE                    | MolecularNet | 1513      | 1     |
|                |         | BBBP                    | MolecularNet | 2039      | 1     |
|                |         | ClinTox                 | MolecularNet | 1478      | 2     |
|                |         | SIDER                   | MolecularNet | 1427      | 27    |
|                |         | Tox21                   | MolecularNet | 7831      | 12    |
|                |         | ToxCast                 | MolecularNet | 8575      | 617   |
| Regression     | RMSE    | ESOL                    | MolecularNet | 1128      | 1     |
|                |         | FreeSolv                | MolecularNet | 642       | 1     |
|                |         | Lipophilicity           | MolecularNet | 4200      | 1     |
|                |         | Caco-2                  | TDC          | 906       | 1     |
|                |         | AqSolDB<br>(Solubility) | TDC          | 9982      | 1     |
|                |         | Acute Toxicity<br>LD50  | TDC          | 7385      | 1     |

**Supplementary Table 9 | Details of p-values in significance test in Figure 3.**

| Metrics        | TamGen                  | GraphBP                 | Pocket2Mol              | TargetDiff              |
|----------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Vina score     | 0.6610                  | 0.2809                  | 1                       | 0.8307                  |
| QED            | 1                       | $1.092 \times 10^{-15}$ | 1                       | $4.153 \times 10^{-5}$  |
| SA score       | $8.221 \times 10^{-25}$ | $3.927 \times 10^{-26}$ | $5.231 \times 10^{-21}$ | $1.401 \times 10^{-10}$ |
| LogP           | 0.9154                  | 1                       | 0.0003                  | $4.525 \times 10^{-5}$  |
| MW             | 0.5616                  | $1.474 \times 10^{-23}$ | $1.058 \times 10^{-5}$  | 0.2761                  |
| TPSA           | 1                       | $3.632 \times 10^{-26}$ | 0.0002                  | 1                       |
| Torsion angles | 0.0003                  | -                       | 0.0008                  | 0.0011                  |

Supplementary Table 10 | Properties of the generated molecules by models in different sampling strategy.

| Metric            | Token-Mol | Token-Mol Topk | TamGen |
|-------------------|-----------|----------------|--------|
| Valid             | 0.973     | 0.976          | 0.997  |
| IntDiv            | 0.849     | 0.841          | 0.829  |
| #Circle           | 52.698    | 44.421         | 38.573 |
| Simi Ori.         | 0.112     | 0.112          | 0.109  |
| Simi Training set | 0.120     | 0.121          | 0.123  |
| Higher Score      | 0.472     | 0.398          | 0.450  |

Supplementary Table 11 | Structural information of generated molecules from models in different sampling strategy.

| Metric                   | Token-Mol | Token-Mol (Topk) | TamGen |
|--------------------------|-----------|------------------|--------|
| Chiral centers           | 1.273     | 1.215            | 1.011  |
| Torsion angles           | 6.386     | 6.327            | 5.118  |
| Rotatable bonds          | 5.326     | 5.345            | 4.225  |
| Ring numbers             | 2.846     | 2.851            | 2.887  |
| Largest ring             | 5.573     | 5.511            | 5.883  |
| Extremely large ring (%) | 0.775     | 0.469            | 0.404  |
| Fused rings (%)          | 47.830    | 48.919           | 47.604 |
| Fused system size        | 2.314     | 2.313            | 2.338  |
| Total fragments          | 21,408    | 18,878           | 18,395 |

Supplementary Table 12 | Model size of Token-Mol and competing/baseline models.

| Total parameters of Token-Mol: 89,788,800 (~89.8M) |       |                     |       |                         |      |
|----------------------------------------------------|-------|---------------------|-------|-------------------------|------|
| Conformation generation                            |       | Property prediction |       | Pocket-based generation |      |
| CGCF                                               | 97K   | XGBoost             | -     | TamGen                  | 112M |
| ConfVAE                                            | 107K  | Chemprop            | 355K  | GraphBP                 | 104K |
| GeoMol                                             | 209K  | MapLight+GNN        | -     | Pocket2Mol              | 3.7M |
| ConfGF                                             | 808K  | GEM                 | 107K  | TargetDiff              | 2.8M |
| GeoDiff                                            | 804K  | K-Bert              | 43.3M |                         |      |
| Tora3D                                             | 30.1M | RT                  | 27.5M |                         |      |

Supplementary Table 13 | Generation speed between different models on Tesla V100 GPU.

|                             | Token-Mol | TamGen | GraphBP | Pocket2Mol | TargetDiff |
|-----------------------------|-----------|--------|---------|------------|------------|
| Times Spent<br>(s/molecule) | 0.365     | 0.120  | 0.750   | 13.103     | 12.971     |

**Supplementary Table 14 | Details of p-values in significance test in Supplementary Figure 9.**

| Metrics    | Token-Mol topk          | TamGen                  |
|------------|-------------------------|-------------------------|
| Vina score | 1                       | 0.3305                  |
| QED        | 1                       | 0.7796                  |
| SA score   | $6.289 \times 10^{-26}$ | $4.110 \times 10^{-25}$ |
| LogP       | 0.0011                  | 0.4577                  |
| MW         | 1                       | 0.2807                  |
| TPSA       | 0.7368                  | 1                       |

**Supplementary Table 15 | The ratio of the potent drug-like molecules generated by each model for each target.**

| Potent drug-like molecules (%)   |               |               |         |               |            |
|----------------------------------|---------------|---------------|---------|---------------|------------|
| Target                           | Token-Mol     | TamGen        | GraphBP | Pocket2Mol    | TargetDiff |
| CDK2                             | <b>9.091</b>  | 6.250         | 0       | 0             | 2.174      |
| AKT1                             | 24.138        | <b>44.330</b> | 1.176   | 0             | 2.174      |
| EGFR                             | 12.037        | <b>32.927</b> | 0       | 0             | 0          |
| DDR1                             | 3.704         | <b>10.588</b> | 0       | 0             | 5.128      |
| ARA2A                            | <b>15.455</b> | 0             | 1.176   | 0             | 14.737     |
| ADRB2                            | 13.889        | <b>26.744</b> | 1.136   | 0             | 1.099      |
| 3CLPro                           | <b>25.225</b> | 2.105         | 1.205   | 0             | 6.410      |
| PD-L1                            | 59.259        | 60.000        | 26.087  | <b>85.841</b> | 9.589      |
| Mean                             | 20.350        | <b>22.868</b> | 3.848   | 10.730        | 5.164      |
| Pearson's r to pIC <sub>50</sub> | 0.413         | 0.814         | 0.321   | 0.300         | -0.362     |

Bold formatting represents the best.

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